

**OPERATIONS TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Pressurizer Pressure and Level		
TASK:	TCAF Pressurizer Pressure Malfunction (Failed open Pressurizer Spray Valve)		
TASK NUMBER:	1140240401		
JPM NUMBER:	11-01 ILOT NRC Sim c		
ALTERNATE PATH:	☒	K/A NUMBER:	010 A4.01
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator		
REFERENCES:	S2.OP-AB.PZR-0001, Rev. 18 (Rev. checked 08-21-12-12)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	4 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	04-12-12
Validated By:	Mulford SME or Instructor	Date:	08-24-12
Approved By:	Training Department	Date:	
Approved By:	Operations Department	Date:	
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

NAME: _____

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**TASK
NUMBER:** 114 024 04 01

SIMULATOR SETUP: IC-253 4% power, BOL.

I/O B304 OVLO Pressurizer Relief Valve Outlet Temperature- Value: 98.6

MALF: PR018A PZR PORV 2PR1 develops leak – Value:40000 Tied to ET-1

RT-1 **PR0019A** PZR Spray Valve 2PS1 fails open

ET-1 kal06tph 23 RCP Stop

**INITIAL
CONDITIONS:**

Reactor power is stable at 4.0% power. The power ascension is on hold temporarily.

INITIATING CUE:

You are the Reactor Operator . Respond to all indications and alarms.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Trip Rx.
2. Stops 21 and 23 RCP.
3. Stops 22 OR 24 RCP.

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	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Simulator Operator: Insert RT-1 on direction of evaluator. MALF: PR0019A , 2PS1 fails open	Note: The time between the 2PS1 failing open and RCS pressure reaching 2,000 psig is ~ 4 minutes and 15 seconds.		
			Recognizes lowering PZR Pressure and/or alarm and/or change in 2PS1 position. Acknowledges OHA E-28 PZR HTR ON PRESS LO. Recognizes 22 PZR B/U heaters energized. If operator refers to ARP for OHA E-28, it directs entry into S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction.		
			Enters S2.OP-AB.PZR-0001 directly or via OHA E-28 ARP. Note: It is acceptable to attempt closing 2PS1 prior to entering S2.OP-AB.PZR-001.		

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	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	3.1	INITIATE Attachment 1 Continuous Action Summary.	Initiates Attachment 1 Continuous Action Summary. Note: There is a CAS action to trip the Rx if RCS pressure lowers to 2,000 psig and continues to drop. IF the operator were to trip the Rx during this JPM based on that CAS action and not at step 3.24.A on page 6 as expected, THEN AFTER the operator has tripped the Rx, verified the Main Turbine is tripped, verified SI not actuated or required and all 4KV vital buses energized, the following cue must be given: Cue: The immediate actions of TRIP-1 have been verified by the CRS and RO. The CRS directs you to continue actions of S2.OP-AB.PZR-0001, while the remaining crew members will continue to perform TRIP-1.		
	3.2	Is POPS in service?	Determines POPS is not in service by initial conditions or console indications.		

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	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	3.3	Is the controlling Pressurizer Pressure Control Channel (I or III) failed?	Determines Pressurizer Pressure Control Channel (I or III) is not failed and GOES TO Step 3.11		
	3.11	Is the Master Pressure Controller failed? (Refer to Attachment 2 for guidance)	Determines Master Pressure Controller is not failed and GOES TO Step 3.17 (may not refer to Attachment 2 if 2PS1 has been noted open with pressure below closing setpoint.)		
	3.17	Is a Spray Valve(s) failed? (Refer to Attachment 2 for guidance)	Determines 2PS1 is failed open.		
	3.18	PLACE the Spray Valve(s) in MANUAL	Depresses MANUAL PB for 2PS1 and verifies AUTO light extinguished and MANUAL light illuminates.		
	3.19	OPERATE the Spray Valves to control pressure consistent with Attachment 2.	Depresses the CLOSE PB for 2PS1 and recognizes that 2PS1 remains open.		
	3.20	PLACE all Pressurizer heaters in MANUAL and ON	Determines all PZR heaters are in MANUAL and ON.		

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	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	3.21	Has pressure control been regained?	Determines pressure control has not been regained since 2PS1 remains open.		
	3.22	Is RCS pressure dropping rapidly?	Determines RCS pressure is dropping rapidly from console indication.		
	3.23	Are Reactor Trip Breakers CLOSED?	Determines Reactor Trip Breakers are closed by initial conditions or console indication.		
*	3.24 A	PERFORM the following: A. TRIP the Reactor.	Initiates a Reactor Trip using either MANUAL TRIP handle.		
	3.24 B	B. Is Reactor Trip Confirmed?	Determines Reactor Trip is confirmed after identifying PRNI reading < 5% and IR NI indication dropping.		
*	3.24 C	STOP 21 <u>AND</u> 23 RCPs:	Depresses STOP PBs for 21 and 23 RCP and verifies start lights extinguish and stop lights illuminates. Simulator Operator: Ensure ET-1 is TRUE when 23 RCP stop PB is depressed.		

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	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	3.24 D	<u>IF</u> Pressurizer Pressure continues to drop, <u>THEN STOP</u> all but one RCP.	Determines Pressurizer Pressure continues to drop, and depresses STOP PB on 22 <u>OR</u> 24 RCP and verifies start light extinguishes and stop light illuminates.		
	3.24 E	<u>GO TO</u> 2-EOP-TRIP-1, Reactor Trip or Safety Injection, <u>AND CONTINUE</u> with this procedure.	Begins performing Immediate Actions of 2-EOP-TRIP-1, Reactor Trip or Safety Injection. Cue: When operator begins performing Immediate Actions of 2-EOP-TRIP-1, Reactor Trip or Safety Injection, state JPM is complete.		

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JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- _____ 1. Task description and number, JPM description and number are identified.
- _____ 2. Knowledge and Abilities (K/A) references are included.
- _____ 3. Performance location specified. (in-plant, control room, or simulator)
- _____ 4. Initial setup conditions are identified.
- _____ 5. Initiating and terminating Cues are properly identified.
- _____ 6. Task standards identified and verified by SME review.
- _____ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- _____ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. _____ Date _____
- _____ 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

Date: _____

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